

Make hot cells larger

Inflators tell the placer that cells in congested GCells should act bigger, encouraging spreading on the next pass

The idea

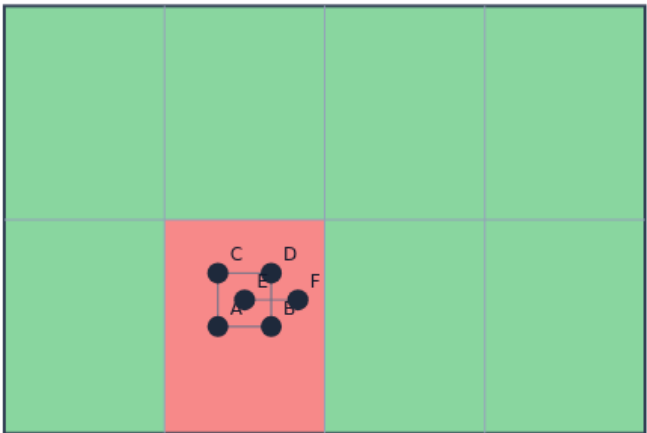
- Map each cell center to a GCell
- If that tile's congestion exceeds one, inflate; otherwise leave width alone
- Alpha around zero point five keeps the demo visible without exploding geometry
- Coordinates stay put, this lab changes widths, not x y

Why inflate

CELL INFLATOR

Why inflate

Step 1 / 5 · idea · module03-01-cell-inflator



Explanation

Make cells in hot GCells act larger so the next place pass spreads.

- Width scale

Coords unchanged here

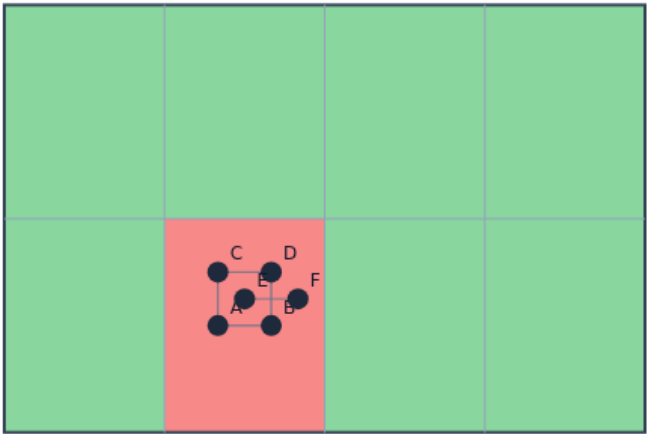
Why inflate

Scale rule

CELL INFLATOR

Scale rule

Step 2 / 5 · rule · module03-01-cell-inflator



Explanation

If $\text{cong} > 1$: $w' = w \cdot (1 + \alpha(c-1))$, $\alpha=0.5$.

- Else keep w

Center cells grow

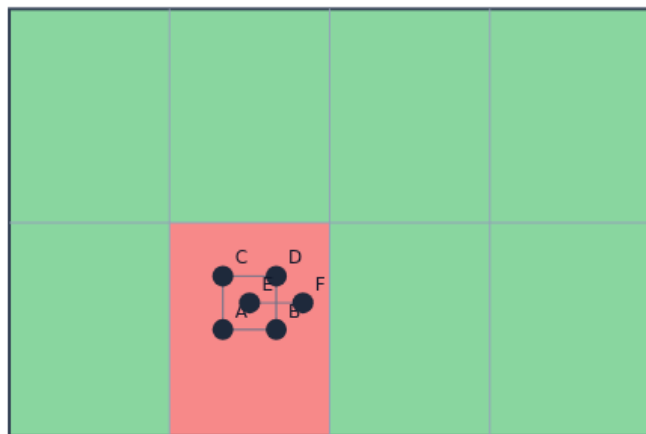
Scale rule

Apply once

CELL INFLATOR

Apply once

Step 3 / 5 · run · module03-01-cell-inflator



Explanation

Compute congestion from RUDY, then inflate widths once.

- Don't double-apply

Reset to base widths

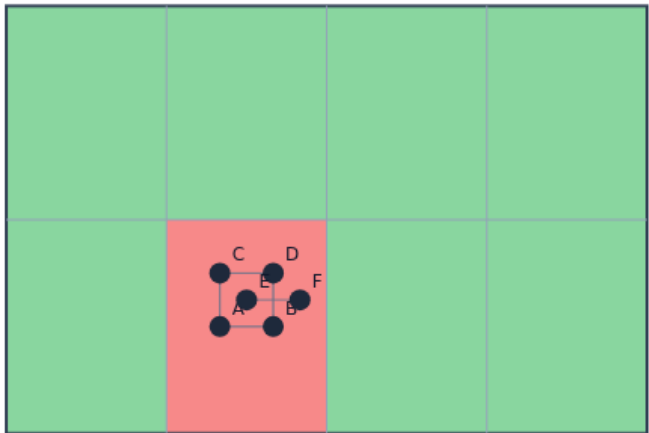
Apply once

Quiet tiles

CELL INFLATOR

Quiet tiles

Step 4 / 5 · quiet · module03-01-cell-inflator



Explanation

Cells in tiles with $\text{cong} \leq 1$ stay at base width.

- Selective

E may stay 1

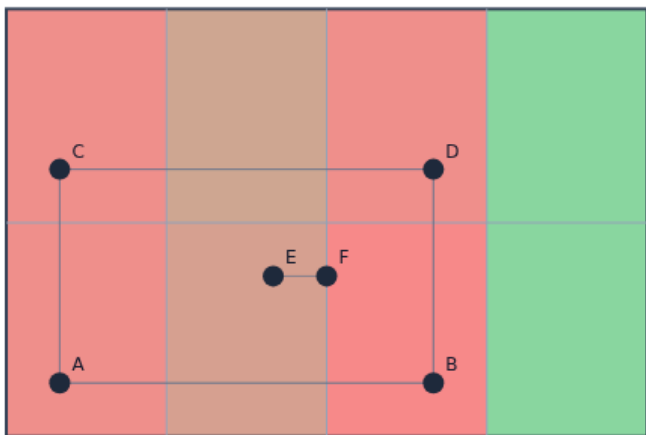
Quiet tiles

Link to place

CELL INFLATOR

Link to place

Step 5 / 5 · link · module03-01-cell-inflator



Explanation

Widths feed the next placer—estimation alone is not enough.

- Feedback course arc

Net weights next

Link to place

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Cell inflation

INTERACTIVE TOOL

Cell inflation

Inflate widths for cells sitting in oversubscribed GCells.

Your job: click *Run inflate* then verify some widths grew where congestion>1. Moving cells changes which tiles are hot.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntro] Inflate someone

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Run feedback

Run inflate

Graph

Reference / metrics

selected: A @ GCell (1,0)

HPWL: 7.0

overflow total: 5.00 · max: 5.00 · count: 1

capacity: 2

view: your placement

inflated: (none)

Browser lab starter

learn_congestion — 01 cell inflator

Implement track

- Implement ``inflate_cells`` with `alpha` equals zero point five
- Print before/after widths for cells sitting in oversubscribed tiles

Pitfalls

- Inflating every cell when only some GCells are hot
- Inflating height when the placer model only tracks width
- Applying inflation twice without resetting to base widths

Your turn

- Clear the inflator challenges
- Next: net weighting, the wirelength-side knob